

- 9 Nina walks at an average speed of 5 km/h, correct to the nearest km/h. She walks for exactly 2 hours.

Work out the lower bound for the distance Nina walks.

$$S = \frac{d}{t} \quad = 4.5 \times 2$$

$$= 9 \text{ km}$$

$$S \times t = d$$

$$\text{Speed} \% \quad \underline{4.5} \quad 5 \quad 5.5$$

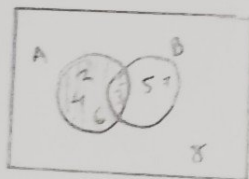
$$\text{Time} \% \quad 2 \text{ hours}$$

$$\underline{9} \text{ km [2]}$$

- 10 $\mathcal{E} = \{n: n \text{ is an integer and } 1 \leq n \leq 8\}$
 $A = \{\text{factors of } 12\} = 2, 3, 4, 6$
 $B = \{\text{odd numbers}\} = 1, 3, 5, 7$

Find

- (a) $A \cap B$



- (b) $n(A' \cup B)$.

$$A \cap B = \{ \underline{3} \} \text{ [1]}$$

$$\underline{7} \quad \underline{3} \text{ [1]}$$

- 11 Write $0.\dot{2}\dot{4}$ as a fraction in its simplest form.

$$100x \left(\begin{array}{l} r = 0.\dot{2}\dot{4} \\ 100r = 24.\dot{2}\dot{4} \end{array} \right) \times 100$$

$$99r = 24$$

$$r = \frac{24}{99} = \frac{8}{33}$$

$$100r = 24.\dot{2}\dot{4}$$

$$r = 0.\dot{2}\dot{4}$$

$$\underline{\frac{8}{33}} \text{ [2]}$$